

Photosynthesis Overview

Photosynthesis is the process plants use to convert light energy into chemical energy. In green leaves, chlorophyll captures sunlight and helps power reactions that turn carbon dioxide and water into glucose.

The light-dependent reactions happen in the thylakoid membranes of chloroplasts. These reactions split water molecules, release oxygen, and store energy in molecules that can support the next stage.

The Calvin cycle uses stored chemical energy to build sugars from carbon dioxide. The sugars can be used for growth, stored as starch, or moved through the plant to support cells that do not photosynthesize.